

Learning When Black-Hole Hair Is Observable: Physical Identifiability, Generalization, and Observable Complementarity

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August 2026

Abstract

High interpolation accuracy does not establish that physical parameters are identifiable. We study this distinction in a controlled Kiselev black-hole benchmark with a validated timelike-emitter and three-dimensional null-geodesic shooting calculation. The forward analysis shows that ringdown is informative for the deformation amplitude k , whereas photon geometry supplies an independent response direction that resolves much of the state-parameter w_0 degeneracy. Relative to ringdown alone, ringdown plus photon geometry improves the grid-median minimum singular value by $4.54\times$ and the median Jacobian condition number by $2.28\times$. In grouped physical interpolation, the corresponding identifiable- w_0 normalized mean absolute error decreases from 0.309 for ringdown to 0.087 for the combined features. Random splitting is more optimistic, while four directional extrapolation tests are substantially harder and split-conformal coverage falls under distribution shift. A uniform 161-phase audit gives a typical normalized prediction shift of 9.48×10^{-4} but exposes catastrophic extrapolation tails. A subsequent targeted 321-phase audit resolves the remaining forward-feature comparisons, yet frozen HGB and random-forest tail shifts exceed predeclared robustness thresholds. Thus the physical complementarity result is supported, but inverse estimator robustness remains conditional. This study is a theoretical identifiability benchmark, not an observational constraint.

1 Introduction

Scientific inverse models can predict accurately for the wrong reason. Repeated realizations of one physical system can leak across random partitions; parameters can be locally degenerate even when observables respond strongly; and a model can interpolate well while failing under extrapolation or small numerical perturbations. These distinctions matter in black-hole spectroscopy, where quasinormal frequencies encode compact-object properties but resolving multiple physical effects remains difficult [8, 2, 3].

Kiselev spacetimes provide a controlled two-parameter

setting. They do not stand in for a detector likelihood or a generic beyond-Kerr theory. Their value here is methodological: exact loss of w_0 dependence at $k = 0$ supplies a known rank-deficient boundary, while nonzero k permits direct comparison of forward Jacobian geometry and inverse-learning behavior [12].

The study complements rapid gravitational-wave inference based on neural posterior estimation and normalizing flows [7, 11, 10, 6, 16]. Those methods address posterior computation; the present question is whether the selected forward observables distinguish the parameters in the first place. This distinction parallels structural and practical identifiability analysis in dynamical inverse problems [19, 21].

This work makes four contributions. It (i) evaluates inverse models with leakage-aware random, grouped, and directional protocols; (ii) validates timelike and null Kiselev geodesic shooting with explicit failure accounting; (iii) quantifies physical Jacobian conditioning and observable complementarity; and (iv) audits uncertainty, noise, learning curves, and 81–161–321 phase-resolution robustness under frozen estimators.

The central narrative rests on three distinctions: sensitivity is not identifiability; interpolation is not generalization, and forward convergence is not estimator robustness.

2 Physical Model and Identifiability Framework

The static spherical line element is

$$ds^2 = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2 d\Omega^2, \quad (1)$$

$$f(r) = 1 - \frac{2M}{r} - \frac{k}{r^{1+3\alpha}}, \quad (2)$$

We use geometrized units with $M = 1$. The dense historical synthetic domain and the later 121-point physical shooting domain are distinct datasets and are not mixed. Leading-eikonal ringdown features use the unstable circular null orbit, with the known limitations of the geodesic-QNM correspondence kept explicit [5, 14, 23].

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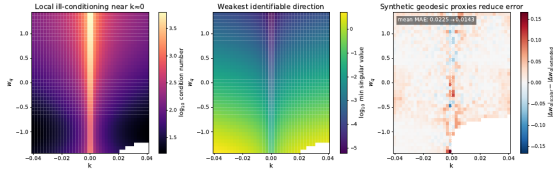


Figure 1: Historical analytic identifiability lens motivating the physical-shooting study. Left: local Jacobian condition number, which grows near $k = 0$. Center: minimum singular value, showing the weakest identifiable direction. Right: change in identifiable- w_q error after adding the historical synthetic geodesic proxies; the mean MAE changes from 0.0225 to 0.0143. White regions are invalid or missing source-grid points and are not interpolated. The first two panels diagnose inverse-map geometry; the third records the original proxy experiment and must not be interpreted as physical ray tracing or geodesic shooting. Proxy augmentation does not resolve the exact $k = 0$ rank loss.

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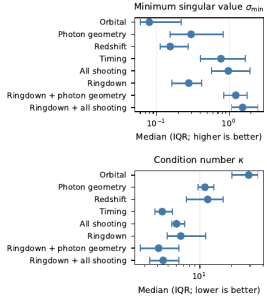


Figure 7: Grid-wide median and interquartile summaries of $\sigma_{\min}$ and condition number. Photon geometry provides the clearest complement to ringdown.

Normalized mean absolute error divides MAE by the full target span. Split-conformal intervals target 90% coverage and are interpreted under their exchangeability assumptions [22, 20, 1]. Feature-noise scales and ev-

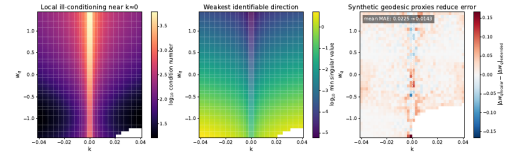


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Photons are propagated in Cartesian spatial coordinates with covariant momentum $\mathbf{p}$ and $p_t = -1$,

$$H_\gamma = \frac{1}{2}[-1/f + \mathbf{p}^2 + (f-1)(\mathbf{n} \cdot \mathbf{p})^2] = 0. \quad (5)$$

Two launch angles are solved so the direct branch reaches the observer plane. The implementation reports failed phases rather than interpolating them or substituting synthetic geodesic proxies. Photon momenta are projected onto the static observer tetrad before sky angles are formed. Redshift uses $1+z = \omega_{\text{emit}}/\omega_{\text{obs}}$ and direct arrival time is compared with $\int (1+z) d\tau_{\text{emit}}$ after removing one common additive constant.

In the Schwarzschild benchmark, all 161 phases succeed for both tested nominal w_q values, which agree because $k = 0$. The independent radial benchmark validates the turning-point period and apsidal advance. The physical 121-point grid also preserves the Hamiltonian constraints and exact rank-loss boundary. The overlapping arrival curves and their numerical residuals are retained as supporting evidence in Appendix G and Fig. 14.

Figure 2 makes the direct-branch geometry explicit using actual validated emitter states, launch angles, photon integrations, and observer geometry. It is a controlled benchmark visualization, not a radiative-transfer or detector simulation.

4 Physical Observable Complementarity

A large response can reveal that the spacetime changed without revealing which parameter caused the change. Timing and redshift responses are often large but nearly collinear in parameter space. Photon geometry rotates the sensitivity direction and therefore contributes information that ringdown lacks. Figure 3 first shows how the same three validated spacetimes produce distinct phase-resolved response patterns.

The standardized local comparison is shown in Fig. 4. The full observer tracks nearly overlap, as expected for these small deformations. Figure 5 therefore retains the physical track as the primary object and shows unnormalized, pointwise matched-phase residuals relative to Schwarzschild. These small resolved changes supply a direction that is

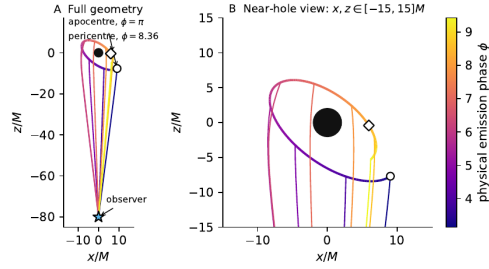


Figure 2: Representative direct-branch null geodesics for $M = 1$, $r_p = 8M$, $r_o = 12M$, $k = 10^{-3}$, $w_q = -0.5$, and observer $(0, 0, -80M)$. Physical phase runs from π at apocentre to 3π ; the diamond marks the numerically detected pericentre and the dark disk is a horizon-scale marker, not a shadow. At each phase, two launch angles are solved until the ray reaches the observer. The right panel shows the stated near-hole coordinate bounds on their actual scale; no trajectory is geometrically magnified. This is a controlled dimensionless benchmark.

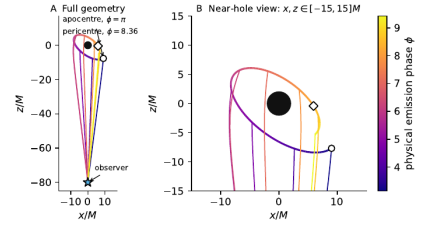


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difficult to see on the full track.

Across the accepted physical grid, ringdown plus photon geometry improves the median $\sigma_{\min}$ by 4.5429 $\times$ and the median condition number by 2.2819 $\times$ relative to ringdown alone. The medians for the combined set are $\sigma_{\min} = 1.2326$ and $\kappa(J) = 4.0731$. Ringdown plus all shooting has $\sigma_{\min} = 1.5312$ and $\kappa(J) = 4.4974$; adding more responsive summaries can raise $\sigma_{\min}$ without producing the best conditioning. Figures 6 and 7 show the pointwise maps and their grid-wide summaries, respectively.

5 Leakage-Aware Inverse-Learning Protocols

Five primary feature sets are ordered consistently: ringdown, photon geometry, all shooting, ringdown plus photon geometry, and ringdown plus all shooting. Here photon geometry and the other integrated quantities are physical shooting observables, distinct from the historical synthetic geodesic proxies. Models are histogram gradient boosting (HGB), random forest (RF), and a target-scaled multilayer perceptron (MLP) [4, 9, 17]. The protocols are random interpolation, two blocked 3×3 grouped physical layouts, and four directional extrapolation tests. All nominal $k = 0$ records are retained and flagged, but ordinary w_q error excludes them because w_q is exactly non-identifiable there.

Normalized mean absolute error divides MAE by the full target span. Split-conformal intervals target 90% coverage and are interpreted under their exchangeability assumptions [22, 20, 1]. Feature-noise scales and every preprocessing transform are learned on training folds. The 81-to-161 and targeted 161-to-321 comparisons apply archived frozen estimators; they do not retrain on higher-resolution test features.

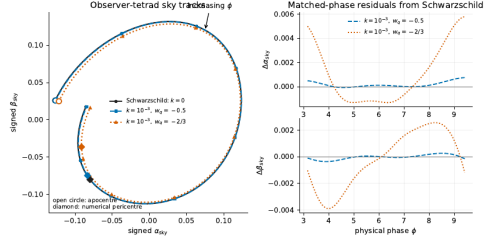


Figure 5: Static-observer-tetrad sky tracks and actual-scale residuals for the same $M = 1$, $r_p = 8M$, $r_a = 12M$, observer $(0, 0, -80M)$ direct-branch benchmark. Open circles mark $\phi = \pi$ apocentre, diamonds mark numerical pericentre, and the arrow gives increasing phase. Residuals are pointwise differences from Schwarzschild at exactly matched successful phases; they are neither normalized for visibility nor interpolated across failures.

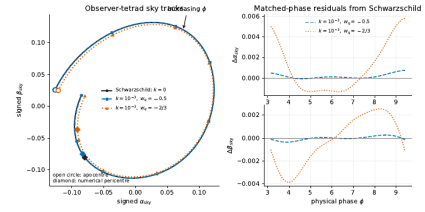


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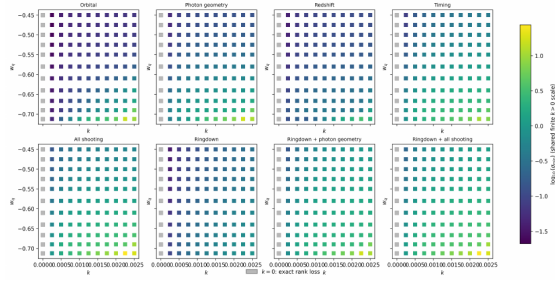


Figure 6: Grid maps of $\log_{10} \sigma_{\min}$ with exact $k = 0$ rank-loss points masked. Missing or invalid values are not interpolated, and all panels use one finite- k color scale.

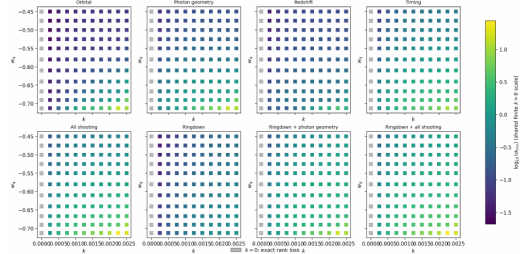


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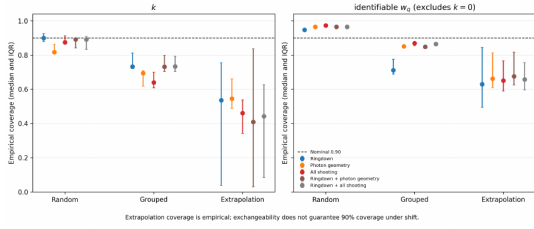


Figure 10: Empirical 90% split-conformal coverage by target and protocol. Ordinary w_q coverage excludes $k = 0$; extrapolation coverage is not guaranteed under distribution shift.

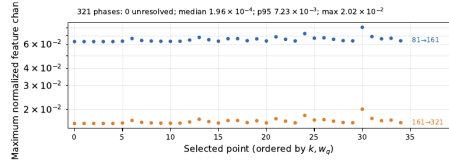


Figure 11: Targeted pointwise maximum normalized feature changes from 81 to 161 and 161 to 321 phases. The 321-phase audit resolves the forward-feature criterion but does not by itself certify inverse estimators.

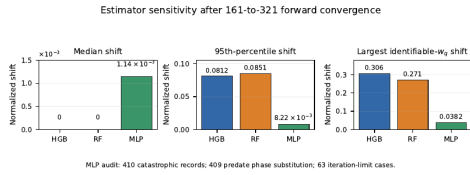


Figure 12: Frozen-estimator sensitivity after forward convergence. Bars show the median, 95th-percentile, and largest identifiable- w_q normalized prediction shift for HGB, RF, and MLP. The audit note summarizes the distinct MLP failure audit. No predictions are clipped.

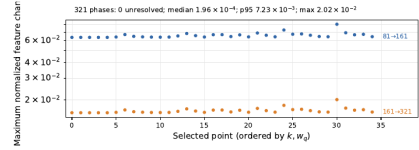


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The targeted 321-phase audit selects 35 points deterministically before reading their 321-phase outcomes. All complete. Median and 95th-percentile normalized 161-to-321 feature changes are 1.963×10^{-4} and 7.234×10^{-3} ; no comparison remains unresolved, ringdown is exactly unchanged, and Jacobian rank is preserved. Forward convergence therefore passes. Figure 11 displays both resolution increments on the same selected-point order.

Frozen trees remain tail-sensitive: HGB and RF 95th-percentile normalized shifts are 0.08122 and 0.08509, and their largest identifiable- w_q shifts are 0.30593 and 0.27053. Their median shifts are zero because many tree predictions do not cross a split threshold. The predeclared perturbation ensembles and fixed Extra Trees smoother do not pass all stability and grouped-performance criteria. Of 410 catastrophic MLP records, 409 are already catastrophic at 161 phases, none is materially worsened by the 321-phase substitution, 63 reach the iteration limit, and no numerical overflow is detected. The decision is Outcome B: the forward map converges, but estimator robustness does not.

Typical predictions are stable, but the tails are not. The earlier MLP maxima are not representative of every estimator, while the targeted tree failures show that the remaining issue also cannot be attributed solely to the MLP. Journal readiness remains conditional.

Figure 12 separates the estimator mechanisms. Tree medians remain exactly unchanged because most perturbations do not cross a split, yet their tails do. The MLP reacts more smoothly to phase substitution, while its catastrophic directional records overwhelmingly predate that substitution.

8 Discussion

Jacobian conditioning answers whether nearby parameter directions are separated by the chosen observables. Training distance answers whether a fitted estimator has support where it is evaluated. Neither statistic subsumes the other. Consequently, $\sigma_{\min}$ can predict broad observable-set ordering without ranking every empirical model or protocol.

Tree ensembles are piecewise and can show zero median response followed by large split-crossing tails. The MLP is smoother under the targeted feature substitution but extrapolates to physically unreasonable values in archived directional tests. These mechanisms are estimator-specific manifestations of a general scientific-ML risk: a converged forward simulator does not ensure a stable inverse surrogate [15, 13, 18].

What is specific to Kiselev is the metric deformation and exact $k = 0$ loss of w_q . What may generalize is the workflow: establish forward rank, seek complementary observables, partition by

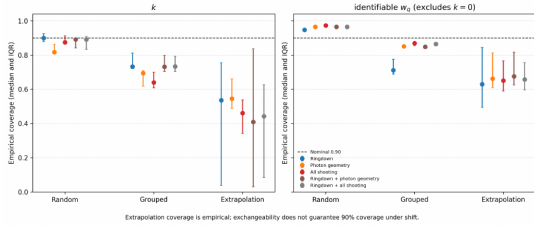


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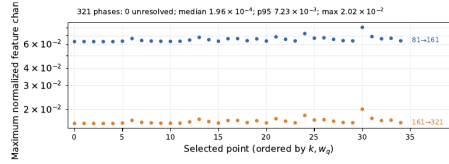


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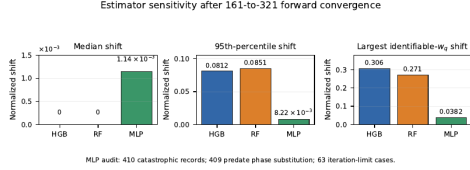


Figure 12: Frozen-estimator sensitivity after forward convergence. Bars show the median, 95th-percentile, and largest identifiable- w_0 normalized prediction shift for HGB, RF, and MLP. The audit note summarizes the distinct MLP failure audit. No predictions are clipped.

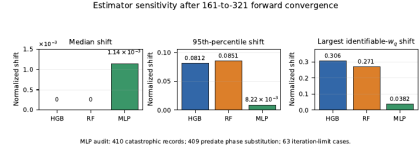


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physical systems, test directional shift, and perturb frozen estimators at independently measured numerical scales.

9 Limitations

The ringdown model is leading-eikonal rather than a dedicated finite- ℓ perturbative spectrum. The spacetime is static. Shooting includes only the direct photon branch, a point emitter and point observer, and no radiative transfer, detector response, secondary images, strain likelihood, calibration error, or model discrepancy. The physical grid is finite. Historical proxy results remain synthetic. Although the targeted audit resolves higher-harmonic forward differences on its selected set, it is not a uniform 321-phase grid and tree tail sensitivity persists. MLP extrapolation is unstable. The verdict is therefore conditional, and no result is an observational constraint.

10 Conclusion

Physical complementarity is supported: ringdown constrains k , while photon geometry supplies an independent direction needed for much of the w_0 recovery. Random interpolation overstates recoverability relative to grouped and directional validation. Finally, numerical and distribution-shift audits are necessary before confident inverse claims: the forward features can converge while fitted estimators remain fragile. These conclusions motivate robust inverse methods and broader physical modeling before observational use.

A Historical Synthetic and Proxy Benchmark

The original dense benchmark used leading-eikonal curves, training-fold PCA, and synthetic geodesic proxies. Waveform-only RF achieved $R^2(k) = 0.8960$ and $R^2(w_0) = 0.9587$; HGB with proxy augmentation achieved 0.9793 and 0.9891. The rank-aware classifier reached macro F1 0.9928 with a 57.6% weak-identifiability fraction. PCA compressed correlated information; it did not create a new physical response direction. These historical results motivated the physical shooting study and are not substituted for it. Figure 13 serves a distinct role from the historical lens in Fig. 1: it compares a learned rank-aware class with the analytic weak-identifiability boundary rather than displaying conditioning or proxy regression-error changes.

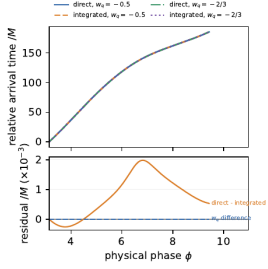


Figure 14: Schwarzschild arrival-time validation. Four arrival-time curves are plotted. The two nominal w_q cases coincide because the $k = 0$ geometry is Schwarzschild, while the direct and integrated-redshift constructions agree within numerical quadrature error. The lower panel shows direct minus integrated arrival time for $w_q = -0.5$ and the direct-arrival difference between $w_q = -0.5$ and $w_q = -2/3$ on their actual numerical scales.

F Noise and Learning Curves

Grouped one-percent feature noise yields ringdown-plus-photon-geometry NMAE 0.0699/0.0853 for k/w_q . Learning curves are still improving at 78 training points, so data limitation remains part of the interpretation.

G Numerical-Validation Thresholds

The 35-point 321-phase run takes 5318.2 s. Maximum hit, timelike-constraint, null-constraint, and impact-parameter drift errors are respectively 7.0913×10^{-8} , 3.1303×10^{-13} , 7.6214×10^{-12} , and 1.7932×10^{-11} . The Schwarzschild arrival-time comparison in Fig. 14 separates the visually overlapping curves from their actual-scale residuals.

Matched Schwarzschild and Kiselev direct rays are compared on their actual coordinate scale in Fig. 15.

Table 2: Targeted 321-phase frozen-estimator robustness.

Model	median	95th percentile	max. w_q
HGB	0	0.08122	0.30593
RF	0	0.08509	0.27053
MLP	0.001143	0.008223	0.03819

H Uniform 161-Phase Convergence Audit

All 121 points complete in 4473.2 s with zero failed points. The audit preserves the exact $k = 0$ boundary and uses no mixed-resolution row, interpolation, or proxy substitution.

I MLP Extrapolation-Failure Audit

The machine-readable table contains standardized input magnitude, hidden activation magnitude, iteration status, overflow flag, training distance, and resolution response for every catastrophic record. It distinguishes optimization failure from extrapolation instability without clipping.

Exact frozen-estimator shift summaries used in Fig. 12 are retained in Table 2 for numerical reference.

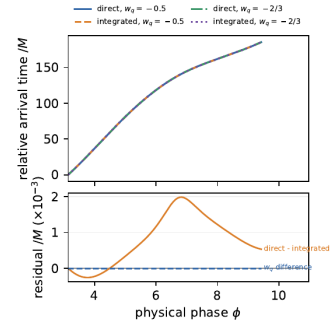


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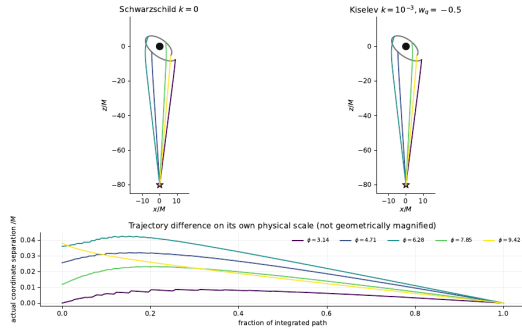


Figure 15: Matched-phase direct rays for Schwarzschild and $k = 10^{-3}$, $w_q = -0.5$ with $M = 1$, $r_p = 8M$, $r_o = 12M$, and observer $(0, 0, -80M)$. The lower panel gives physical coordinate separation versus integrated path fraction. Differences are shown on their actual scale and are not geometrically magnified.

References

- [1] Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification, 2021.
- [2] Emanuele Berti, Vitor Cardoso, and Clifford M. Will. Gravitational-wave spectroscopy of massive black holes with the space interferometer lisa. *Physical Review D*, 73:064030, 2006.
- [3] Swetha Bhagwat, Xisco Jimenez Forteza, Paolo Pani, and Valeria Ferrari. Ringdown overtones, black hole spectroscopy, and no-hair theorem tests. *Physical Review D*, 101:044033, 2020.
- [4] Leo Breiman. Random forests. *Machine Learning*, 45:5–32, 2001.
- [5] Vitor Cardoso, Alex S. Miranda, Emanuele Berti, Helvi Witte, and Vilson T. Zanchin. Geodesic stability, lyapunov exponents and quasinormal modes. *Physical Review D*, 79:064016, 2009.
- [6] Kyle Cramer, Johann Brehmer, and Gilles Louppe. The frontier of simulation-based inference. *Proceedings of the National Academy of Sciences*, 117:30055–30062, 2020.
- [7] Maximilian Dax, Stephen R. Green, Jonathan Gair, Jakob H. Macke, Alessandra Buonanno, and Bernhard Scholkopf. Real-time gravitational-wave science with neural posterior estimation. *Physical Review Letters*, 127:241103, 2021.
- [8] Olaf Dreyer, Bernard Kelly, Badri Krishnan, Lee Samuel Finn, David Garrison, and Ramon Lopez-Aleman. Black-hole spectroscopy: testing general relativity through gravitational-wave observations. *Classical and Quantum Gravity*, 21:787–804, 2004.
- [9] Jerome H. Friedman. Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29:1189–1232, 2001.
- [10] Hunter Gabbard, Chris Messenger, Ik Siang Heng, Francesco Tonolini, and Roderick Murray-Smith. Bayesian parameter estimation using conditional variational autoencoders for gravitational-wave astronomy. *Nature Physics*, 18:112–117, 2022.
- [11] Stephen R. Green, Christine Simpson, and Jonathan Gair. Gravitational-wave parameter estimation with autoregressive neural network flows. *Physical Review D*, 102:104057, 2020.
- [12] V. V. Kiselev. Quintessence and black holes. *Classical and Quantum Gravity*, 20:1187–1198, 2003.

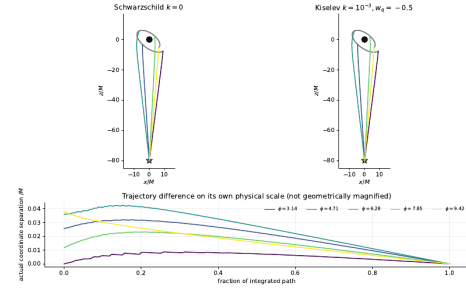


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Exact frozen-estimator shift summaries used in Fig. 12 are retained in Table 2 for numerical reference.

Table 2: Targeted 321-phase frozen-estimator robustness.

Model	median	95th percentile	max. w_q
HGB	0	0.08122	0.30593
RF	0	0.08509	0.27053
MLP	0.001143	0.008223	0.03819

References

- [1] Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification, 2021.